Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

FIFTH SEMESTER BTECH. (E & C) DEGREE END SEMESTER EXAMINATION
JANUARY/FEBRAURY 2021
SUBJECT: INTRODUCTION TO COMMUNICATION SYSTEMS (ECE - 4304)

TIME: 3 HOURS **MAX. MARKS: 50**

Instructions to candidates
- Answer **ALL** questions.
- Missing data may be suitably assumed.

1A. Why synchronization is critical in a communication system. Design a Digital communication system with the help of a neat block diagram showing how synchronization leads to an acceptable bit error rate.

1B. With a proper tabular column, list down the frequency, wavelength and applications pertaining to the Electromagnetic spectrum from 30Hz to 300GHz.

1C. A system has power amplifier with a 60 dB gain, and has an output power of 10 W. What is the input power that is required to feed the antenna? If you want a further improvement in bit error rate for a system, what is the technique that you employ, Comment on the results obtained.

(4+3+3)

2A. Consider an Armored outdoor fibre optic cable, how exactly you can choose a loose tube structure in the cable which also avoids water blocking. Explain with a relevant diagram.

2B. Two silica fibres that are doped with 18 percent and 8 percent mole fractions of GeO_2 respectively. Compare the ultraviolet absorptions at wavelengths of 0.9 μm and 1.8 μm. Comment on the results.

2C. What is a micro bend in an OFC system? With the help of a neat diagram explain the losses which are induced due to bending.

(4+3+3)

3A. The transmit power of a certain system is 10 watts, and given that both transmit and receive parabolic antennas have a diameter of 3 m. The antenna efficiency is 55% for both. The satellite is in a GSO location, with a range of 35900 km. The frequency of operation is 12 GHz. Determine the received power. Assume that the gains of Tx and Rx antennas to be 49dBi.

3B. With the generic block diagram, explain a Telemetry Tracking and Commanding system.

3C. The downlink signal received at the earth station is 3 GHz, what local-oscillator frequencies f_{lo} are needed to achieve IFs of 770 and 140 MHz?

(4+3+3)

4A. Consider a FSO system, define how Acquisition Tracking and Pointing happens in a FSO system. How to obtain the first link point from one satellite on LEO to another satellite on GEO.

4B. How handoff takes place in a mobile environment, define how channel allocation strategies help the network to have a proper handoff.

4C. Comment on the practical limitations of a Nano Cell Zone concept.

(4+3+3)

5A. With the help of a neat block diagram explain the concept of a MTI Doppler Radar. How a system identifies a moving target.

5B. Explain the concept of a bluetooth pico-net with scatter-net link.

(5+5)